PRINCIPLES OF ECONOMICS

15-Page Colourful Revision Note

BRUR CSE | Semester Examination | Based on Mankiw Ch. 4 + Lecture Chapter 2

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HOW TO USE THIS NOTE (3 Days Plan)

- **Day 1:** Pages 2 to 6 (definitions, PPF, circular flow). Draw the PPF and circular flow 3 times from memory.
- **Day 2:** Pages 7 to 13 (demand, supply, equilibrium). Practice the 3-step method on 5 events.
- **Day 3:** Pages 14 and 15 (maths and extras). Solve all numericals without looking.
- **Last hour:** Read only the coloured BOX items on every page.

COLOUR CODE

BLUE = Demand side	**RED** = Supply side	**GREEN** = Equilibrium or efficient	**ORANGE** = Surplus or warning

BIG IDEA: Economics = study of CHOICE under SCARCITY. Every topic in this note comes from three ideas: (1) resources are scarce, (2) choices have opportunity costs, (3) prices (demand and supply) coordinate choices.

TOP 10 THINGS EXAMINERS ALWAYS ASK

- Definitions of Smith, Marshall, Robbins, Samuelson with **criticism**
- Circular-flow diagram (labelled)
- PPF graph, opportunity cost, growth shift
- Positive vs normative statements with examples
- Law of demand and law of supply
- Shifters of demand and supply with examples
- Shift vs movement along the curve
- Equilibrium numerical (Qd = 50 - 5P, Qs = 5 + 10P)
- Surplus and shortage with graph
- 3-step method with combined shifts (ambiguous effect)

Definitions of Economics

Economist	Definition and Focus	Criticism
Adam Smith 1776 (Classical)	"Economics is the **science of wealth**." Focus: production and accumulation of wealth; how nations grow richer. Called **Father of Economics** (book: The Wealth of Nations).	Too narrow. Ignores human and social welfare.
Alfred Marshall 1890 (Neoclassical)	"Economics is the study of mankind in the **ordinary business of life**." Focus: human **welfare**, income, consumption, well-being. A social science.	Limited to material welfare only.
Lionel Robbins 1932 (Scarcity)	"Economics is the science which studies human behavior as a relationship between **ends** and **scarce means** which have **alternative uses**." Emphasis: scarcity, choice, opportunity cost. Covers material and non-material wants.	Ignores human welfare and ethics.
Paul Samuelson Modern	"Study of how people and society choose, **with or without money**, to employ scarce productive resources..." Integrates scarcity, choice, efficiency and welfare. Covers micro and macro, future planning and policy.	Considered the **most complete** definition.
J. M. Keynes 1930s (Macro)	(No direct quote) Focus on **employment, income, output and demand**. Advocated government intervention. Developed macroeconomics; transformed economics after the Great Depression.	Mainly short-run, macro focus.

MEMORY TRICKS

Smith = **W**ealth	**M**arshall = **W**elfare	**R**obbins = **R**arity (Scarcity)	**S**amuelson = Scarce + Social choice

EVOLUTION OF DEFINITIONS (Timeline)

1776 Smith WEALTH	**1890** Marshall WELFARE	**1932** Robbins SCARCITY	**1930s** Keynes MACRO	**Modern** Samuelson COMPLETE

HOW TO WRITE A 10-MARK ANSWER

- **Intro:** one line - meaning of economics changed over time.
- **Body:** five definitions, each with **quote + focus + criticism** (use the table above).
- **Compare:** draw a small table (Economist, Focus, Criticism).
- **Conclusion:** Samuelson gives the most complete modern view; Keynes added macroeconomics.

ONE-LINE REVISION: Smith = Wealth (too narrow), Marshall = Welfare (material only), Robbins = Scarcity (ignores ethics), Samuelson = with or without money (complete), Keynes = Employment and demand (govt. intervention).

Core Concepts

SCARCITY

- **Unlimited wants** vs **limited resources**.
- Root of all economic problems.
- Forces every society to **choose**.
- Every choice has an opportunity cost.

OPPORTUNITY COST

- Value of the **next best alternative** given up.
- Includes time, effort and money.
- Example: 3 hours study = lost part-time wage.
- On PPF: slope = opportunity cost.

FACTORS OF PRODUCTION (Inputs)

Factor	Meaning	Reward	Example
Land	Natural resources	Rent	Farmland, minerals
Labor	Human effort	Wages	Teacher, programmer
Capital	Man-made tools	Interest	Machines, computers
Entrepreneurship	Organizing and risk-taking	Profit	Business owner

INDIVIDUAL TO GLOBAL (Scale of Economics)

Individual | Household | Family | Society | Village/Ward | Upazila-District-Division | Country | Continent | Global

MICROECONOMICS vs MACROECONOMICS

Basis	Microeconomics	Macroeconomics
Definition	How households and firms make decisions and interact in markets	Economy-wide phenomena: inflation, unemployment, growth
Unit of study	Individual consumers, firms, single markets	Whole economy (aggregates)
Central problem	Price determination	Income and employment determination
Also called	Price theory	Income theory
Tools	Demand, supply, cost, market structure	GDP, price level, money supply, policy
Example	Price of rice; a firm output decision	National inflation rate; GDP growth of Bangladesh

THREE BASIC ECONOMIC QUESTIONS

WHAT to produce?
Which goods and how much (cars or computers)

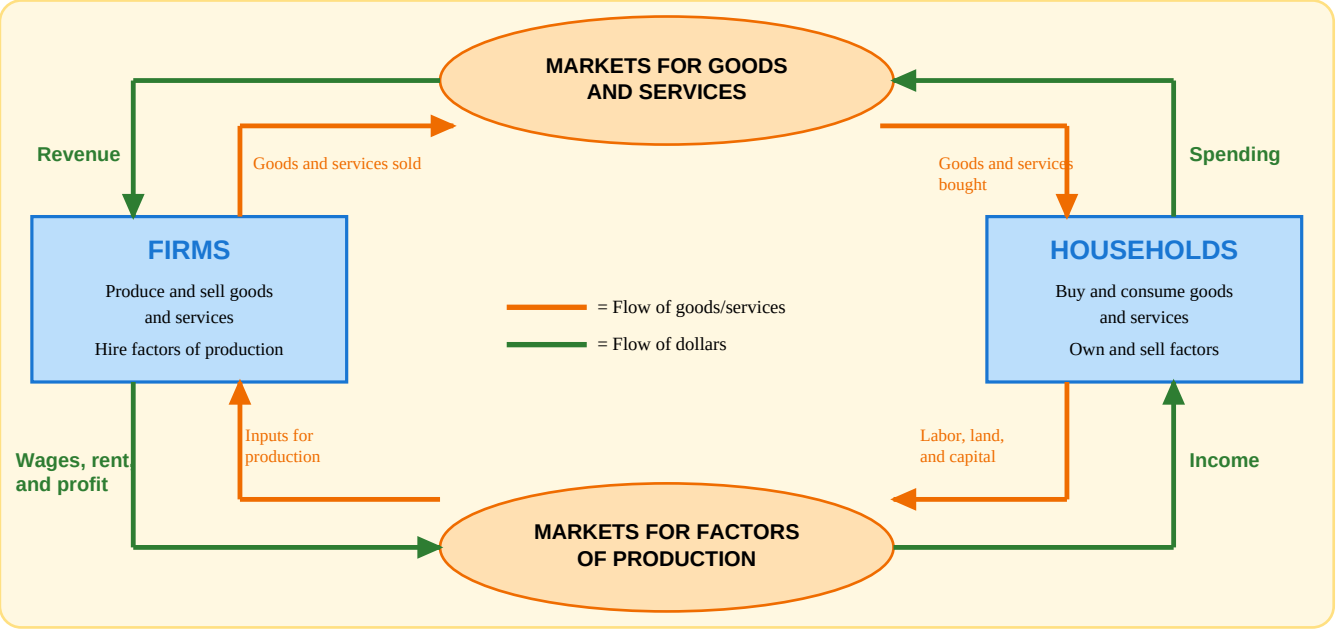
HOW to produce?
Which technique (labor or capital intensive)

FOR WHOM to produce?
How output is shared among people

KEY ASSUMPTION: Ceteris paribus = "other things equal". We change ONE variable (price) and hold the rest constant. If price changes -> movement along the curve. If anything else changes -> the curve shifts.

Circular-Flow Diagram

DEFINITION: A **visual model of the economy** that shows how dollars flow through markets among households and firms.



TWO SECTORS

- **Households:** buy and consume goods and services; own and sell factors of production.
- **Firms:** produce and sell goods and services; hire and use factors of production.

TWO MARKETS

- **Goods and services market:** firms sell, households buy.
- **Factors of production market:** households sell (labor, land, capital), firms buy.

ALL FLOWS AT A GLANCE

Flow	From to To	Type
Spending	Households to goods market	Money (green)
Revenue	Goods market to Firms	Money (green)
Wages, rent, profit	Firms to factor market	Money (green)
Income	Factor market to Households	Money (green)
Goods and services sold / bought	Firms to goods market to Households	Real (orange)
Labor, land, capital / inputs	Households to factor market to Firms	Real (orange)

ADVANTAGES

- Simple way to understand the economy
- Shows interdependence of households and firms
- Shows money and goods move in opposite directions
- Base for GDP and national income

DISADVANTAGES

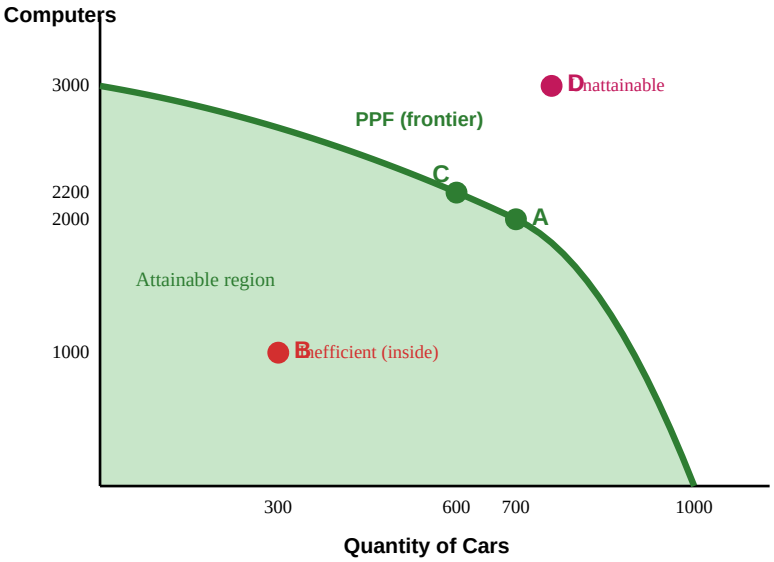
- Ignores government and taxes
- Ignores savings, banks and investment
- Ignores foreign trade
- Assumes all income is spent (unrealistic)

REMEMBER: One person's **spending** is another person's **income**, so money moves in a circle. Orange arrows = goods/services, green arrows = dollars; they run in opposite directions.

Production Possibilities Frontier

DEFINITION: A graph that shows the **combinations of output** the economy can possibly produce given the available **factors of production** and the available **production technology**.

Point	Computers	Cars	Status
(axis)	0	1000	On frontier
A	2000	700	Efficient
C	2200	600	Efficient
(axis)	3000	0	On frontier
B	1000	300	Inefficient (inside)
D	?	?	Unattainable (outside)



WHAT EACH REGION MEANS

- **On the PPF (A, C):** efficient, resources fully used
- **Inside (B):** inefficient, resources wasted or unemployed
- **Outside (D):** cannot be produced now
- **Slope:** downward (producing more of one means less of the other)

OPPORTUNITY COST FROM THE PPF

Move	Computers gained	Cars lost	Opportunity cost of 1 computer
(0 comp, 1000 cars) to A	+2000	-300	0.15 car
A to C	+200	-100	**0.50 car** (1 car = 2 computers)
C to (3000 comp, 0 cars)	+800	-600	0.75 car

WHY IS THE PPF BOWED OUTWARD?

- Resources are not equally good at making both goods.
- Law of **increasing opportunity cost**.
- Best-suited resources move first (low cost).
- Later moves use less suitable resources (higher cost).

THE PPF SHOWS 5 CONCEPTS

- **Scarcity:** points outside are unattainable
- **Choice:** pick a point on the frontier
- **Opportunity cost:** the slope
- **Efficiency:** points on the frontier
- **Growth:** shift of the frontier

EXAM EXAMPLE: B (1000 comp, 300 cars) to A (2000 comp, 700 cars) gives **+1000 computers AND +400 cars** with no sacrifice. That is why B is inefficient.

PPF Shift and Positive vs Normative

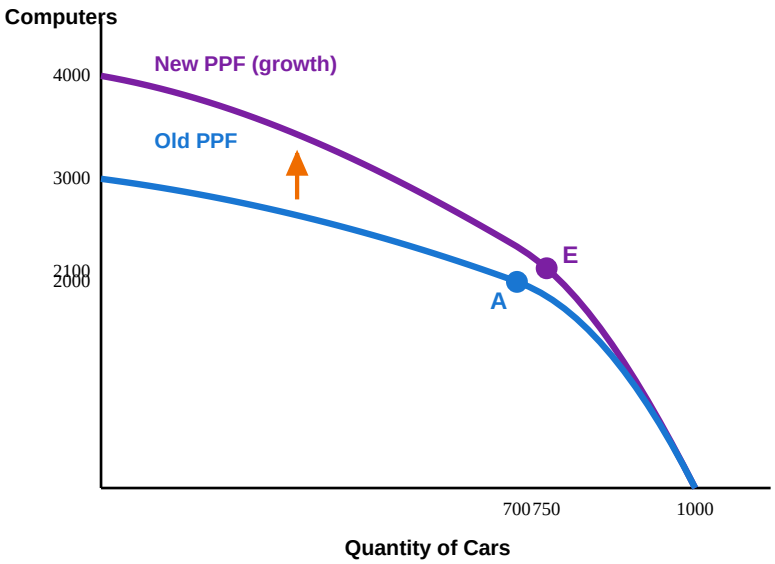
SHIFT OF THE PPF (Economic Growth)

Before: max computers 3,000, max cars 1,000.
Point A: 700 cars, 2,000 computers.

After technology improvement: max computers rise to 4,000; max cars stay 1,000.
Point E: 750 cars, 2,100 computers.

MEANING: E has MORE cars and MORE computers than A. The economy can now produce more of both goods = **growth**.

- **Outward shift:** better technology, more labor or capital, education, new resources.
- **Inward shift:** war, natural disaster, resource depletion, economic collapse.



Outward shift (growth)	Inward shift (decline)
Better technology	War or natural disaster
More labor or capital	Fall in labor or capital
Better education and skills	Resource depletion
Discovery of resources	Economic collapse

POSITIVE vs NORMATIVE STATEMENTS (Perception vs Reality)

Basis	Positive statement	Normative statement
Meaning	Claims that attempt to **describe the world as it is**	Claims that attempt to **prescribe how the world should be**
Nature	Factual, objective	Opinion, value judgment
Proof	Can be tested with data (true or false)	Cannot be proved true or false
Key words	is, are, will	should, ought, must
Example 1	Minimum wage laws cause unemployment.	The government should raise the minimum wage.
Example 2	Inflation in Bangladesh was 9% last year.	Inflation ought to be below 5%.
Branch	Positive economics	Normative economics

CLASSIFY THESE (Practice)

Statement	Type
"A rise in the price of tea reduces the quantity of tea demanded."	**Positive**
"The government should provide free textbooks."	**Normative**
"The unemployment rate is 5 percent."	**Positive**
"Rich people ought to pay higher taxes."	**Normative**

RULE: If you can **test it with data**, it is POSITIVE. If it contains **opinion or "should"**, it is NORMATIVE.

Markets and Demand

MARKET

- A group of **buyers and sellers** of a particular good or service.
- Buyers as a group determine **demand**.
- Sellers as a group determine **supply**.

COMPETITIVE MARKETS

- **Competitive:** many buyers and sellers, each has negligible impact on price.
- **Perfectly competitive:** (1) all goods exactly the same, (2) so many buyers/sellers that no one affects price.
- Everyone is a **price taker**.

DEMAND - KEY DEFINITIONS

- **Quantity demanded (Qd):** amount of a good buyers are **willing and able** to purchase.
- **Law of demand:** other things equal, price up -> Qd down; price down -> Qd up (negative relationship).
- **Demand schedule:** a table of price and quantity demanded. **Demand curve:** its graph, **downward sloping**.

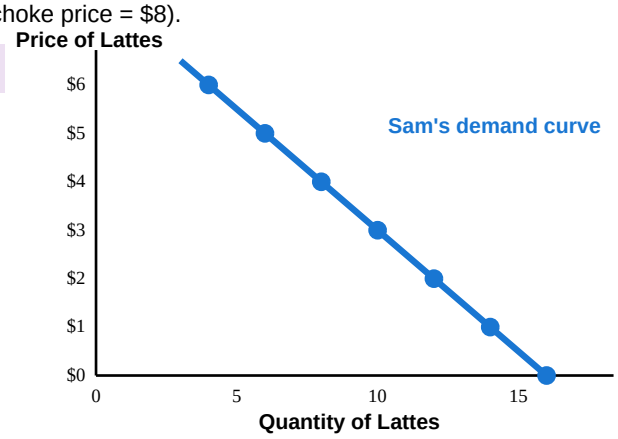
SAM'S DEMAND SCHEDULE (Lattes)

Price	$0	$1	$2	$3	$4	$5	$6
Qty demanded	16	14	12	10	8	6	4

- Each $1 rise in price lowers quantity by 2 -> **Qd = 16 - 2P** (choke price = $8).

WHY DOES DEMAND SLOPE DOWNWARD?

- **Substitution effect:** buyers switch to cheaper substitutes
- **Income effect:** higher price cuts real purchasing power
- **Diminishing marginal utility:** extra units give less satisfaction
- **New buyers** enter at lower prices



MARKET DEMAND = SUM OF INDIVIDUAL DEMANDS

Price	$0	$1	$2	$3	$4	$5	$6
Sam Qd	16	14	12	10	8	6	4
Dean Qd	8	7	6	5	4	3	2
Market Qd	**24**	**21**	**18**	**15**	**12**	**9**	**6**

RULE: Market demand curve = **horizontal sum** of individual curves (add QUANTITIES at each price, never prices). 16-2P + 8-P gives **Qd = 24 - 3P**.

Demand Curve Shifters

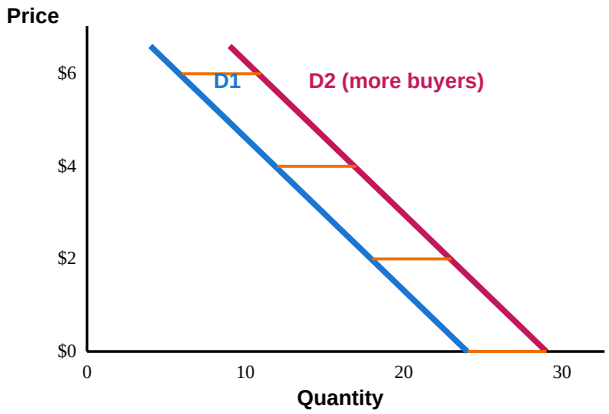
IDEA: The demand curve shows how **price** affects quantity demanded, other things equal. The "other things" are the **non-price determinants**. When they change, the whole curve **SHIFTS** (right = increase in demand, left = decrease).

FIVE SHIFTERS - memory aid: I - P - T - E - N

Shifter	Change	Effect on demand	Example
Income (normal good)	Income up	D shifts RIGHT	Restaurant meals, smartphones
Income (inferior good)	Income up	D shifts LEFT	Instant noodles, public bus
Price of substitute	Price of substitute up	D shifts RIGHT	Pizza price up, hamburger demand up
Price of complement	Price of complement up	D shifts LEFT	Computer price up, software demand down
Tastes	Tastes move toward good	D shifts RIGHT	Atkins diet in the 90s raised egg demand
Expectations	Expect higher income or prices	D shifts RIGHT now	Expect pay rise: more restaurant meals now
Number of buyers	More buyers	D shifts RIGHT	Rise by 5 at each price

NUMBER OF BUYERS (Slide Example)

Price	Old Qd	New Qd
$0	24	29
$1	21	26
$2	18	23
$3	15	20
$4	12	17
$5	9	14
$6	6	11



SUMMARY: VARIABLES THAT INFLUENCE BUYERS

Variable	A change in this variable...
Price of the good itself	**MOVEMENT ALONG** the demand curve
Income, related goods, tastes, expectations, number of buyers	**SHIFTS** the demand curve

QUICK RULES: **Normal good:** income up -> demand up. **Inferior good:** income up -> demand down. **Substitutes:** same direction. **Complements:** opposite direction.

Related Goods and Active Learning 1

SUBSTITUTES

- Price of one up -> demand for the other **UP**.
- Used instead of each other.
- **Example:** pizza and hamburgers (pizza price up -> hamburger D shifts right).
- Others: Coke and Pepsi, laptops and tablets, music CDs and downloads.

COMPLEMENTS

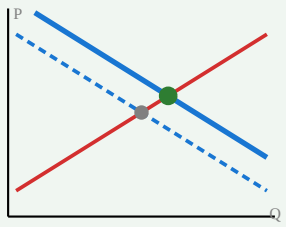
- Price of one up -> demand for the other **DOWN**.
- Used together.
- **Example:** computers and software (computer price up -> software D shifts left).
- Others: tuition and textbooks, bagels and cream cheese, eggs and bacon.

Basis	Substitutes	Complements
Price of A rises	Demand for B rises	Demand for B falls
Cross effect	Positive	Negative
Example	Coke - Pepsi	Car - petrol

ACTIVE LEARNING 1: Demand for MUSIC DOWNLOADS

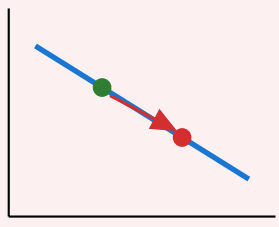
Scenario	Reasoning	Result
A. Price of iPods falls	iPods and downloads are **complements**	**D shifts RIGHT**
B. Price of music downloads falls	Own price changed	**NO shift**; move DOWN along D (lower P, higher Q)
C. Price of music CDs falls	CDs and downloads are **substitutes**	**D shifts LEFT**

A: iPod price falls



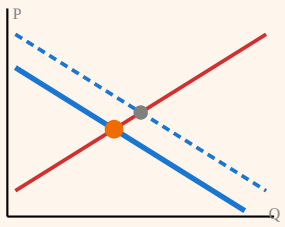
P up | Q up

B: own price falls



MOVEMENT along D

C: CD price falls



P down | Q down

CHANGE IN DEMAND vs CHANGE IN QUANTITY DEMANDED

Basis	Change in DEMAND	Change in QUANTITY DEMANDED
Cause	Non-price determinant (income, tastes, buyers, related goods, expectations)	Change in the good's **own price**
Graph	Whole curve **SHIFTS**	**MOVEMENT ALONG** a fixed curve
Words	Increase / decrease in demand	Expansion / contraction of quantity demanded

GOLDEN RULE: Own price changes = move along the curve. ANYTHING else changes = shift the curve.

Supply

SUPPLY - KEY DEFINITIONS

- **Quantity supplied (Qs):** amount of a good sellers are **willing and able** to sell.
- **Law of supply:** other things equal, price up -> Qs up; price down -> Qs down (positive relationship).
- **Supply schedule:** table of price and quantity supplied. **Supply curve:** its graph, **upward sloping**.

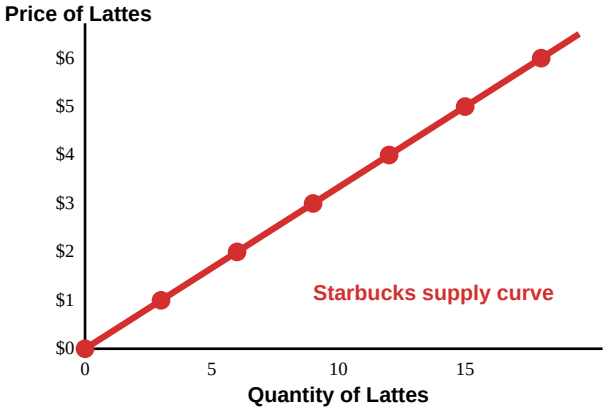
STARBUCKS' SUPPLY SCHEDULE (Lattes)

Price	$0	$1	$2	$3	$4	$5	$6
Qty supplied	0	3	6	9	12	15	18

- Each $1 rise in price adds 3 lattes -> **Qs = 3P**.

WHY DOES SUPPLY SLOPE UPWARD?

- **Profit motive:** higher price = more profit per unit
- **Rising marginal cost:** more output costs more per unit
- **New sellers** are attracted by high price
- **Existing firms** expand production



MARKET SUPPLY = SUM OF INDIVIDUAL SUPPLIES

Price	$0	$1	$2	$3	$4	$5	$6
Starbucks Qs	0	3	6	9	12	15	18
Peet's Qs	0	2	4	6	8	10	12
Market Qs	**0**	**5**	**10**	**15**	**20**	**25**	**30**

RULE: Market supply = **horizontal sum** of individual supply curves. 3P + 2P gives **Qs = 5P**.

DEMAND vs SUPPLY - MASTER COMPARISON

Basis	Demand	Supply
Who	Buyers	Sellers
Law	Price up, Qd down	Price up, Qs up
Slope	Downward	Upward
Shifters (memory)	I-P-T-E-N	I-T-E-N
Function	Qd = a - bP	Qs = c + dP
Example	Sam's lattes	Starbucks' lattes

Supply Curve Shifters

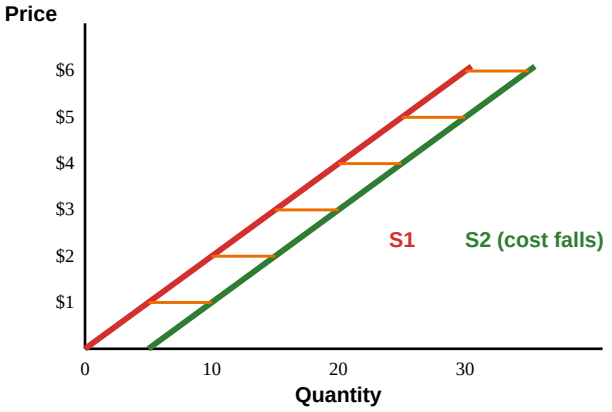
IDEA: The supply curve shows how **price** affects quantity supplied, other things equal. The "other things" are **non-price determinants of supply**. When they change, the curve **SHIFTS**.

FOUR SHIFTERS - memory aid: I - T - E - N

Shifter	Change	Effect on supply	Example
Input prices	Input price falls (supply is negatively related to input price)	S shifts RIGHT	Price of milk falls: more lattes (+5 at each price)
Technology	Cost-saving technology	S shifts RIGHT	New technology cuts hybrid car cost
Expectations	Expect higher future price (non-perishable)	S shifts LEFT now	Texas oil owners hold back inventory
Number of sellers	More sellers	S shifts RIGHT	New firms enter the market

INPUT PRICE FALLS (Slide Example)

Price	Old Qs	New Qs
$1	5	10
$2	10	15
$3	15	20
$4	20	25
$5	25	30
$6	30	35



SUMMARY: VARIABLES THAT INFLUENCE SELLERS

Variable	A change in this variable...
Price of the good itself	**MOVEMENT ALONG** the supply curve
Input prices, technology, expectations, number of sellers	**SHIFTS** the supply curve

ACTIVE LEARNING 2: Supply of TAX RETURN SOFTWARE

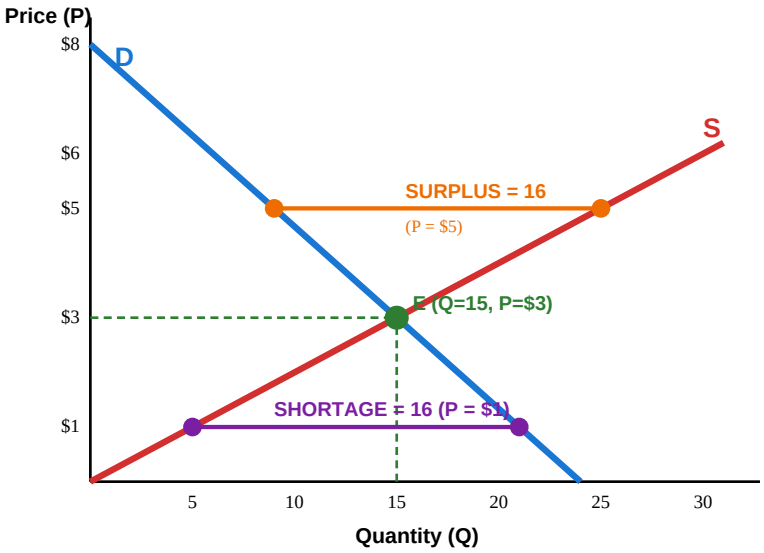
Scenario	Reasoning	Result
A. Retailers cut the price of the software	Own price changed	**NO shift**; move DOWN along S (lower P, lower Q)
B. Technological advance lowers cost	Technology improves	**S shifts RIGHT** (more Q at each price)
C. Professional preparers raise their price	**Trick question!** Preparers are a substitute for the software	Shifts **DEMAND** (right), **not** supply

REMEMBER: **Change in supply** = shift of S (non-price cause). **Change in quantity supplied** = movement along a fixed S (price changes).

Equilibrium, Surplus and Shortage

EQUILIBRIUM

- **Equilibrium:** price has reached the level where quantity supplied = quantity demanded.
- **Equilibrium price:** price where Qs = Qd (market-clearing price).
- **Equilibrium quantity:** Q supplied and demanded at that price.
- **Lattes:** P = **$3**, Q = **15**.



P	$0	$1	$2	$3 (E)	$4	$5	$6
Qd	24	21	18	**15**	12	9	6
Qs	0	5	10	**15**	20	25	30
Result	Shortage 24	Shortage 16	Shortage 8	**Equilibrium**	Surplus 8	Surplus 16	Surplus 24

SURPLUS (Excess Supply)

- **Qs > Qd**, price **above** equilibrium.
- Example: P = $5, Qd = 9, Qs = 25 -> **surplus = 16**.
- **Adjustment:** sellers have unsold stock -> cut price -> Qd rises, Qs falls -> price falls until equilibrium.

SHORTAGE (Excess Demand)

- **Qd > Qs**, price **below** equilibrium.
- Example: P = $1, Qd = 21, Qs = 5 -> **shortage = 16**.
- **Adjustment:** buyers cannot get goods -> sellers raise price -> Qd falls, Qs rises -> price rises until equilibrium.

Basis	Surplus	Shortage
Also called	Excess supply	Excess demand
Condition	Qs > Qd	Qd > Qs
Price level	Above equilibrium	Below equilibrium
Pressure on price	Downward	Upward
Example (slides)	P = $5: surplus 16	P = $1: shortage 16

HOW PRICES ALLOCATE RESOURCES

- "Markets are usually a good way to organize economic activity."
- In market economies, prices adjust to balance supply and demand.
- Equilibrium prices are **signals** that guide economic decisions and allocate scarce resources.
- Equilibrium is **stable**: any deviation triggers forces that restore it.

REMEMBER: Price above E -> SURPLUS -> price falls. Price below E -> SHORTAGE -> price rises. Markets self-correct until **Qs = Qd**.

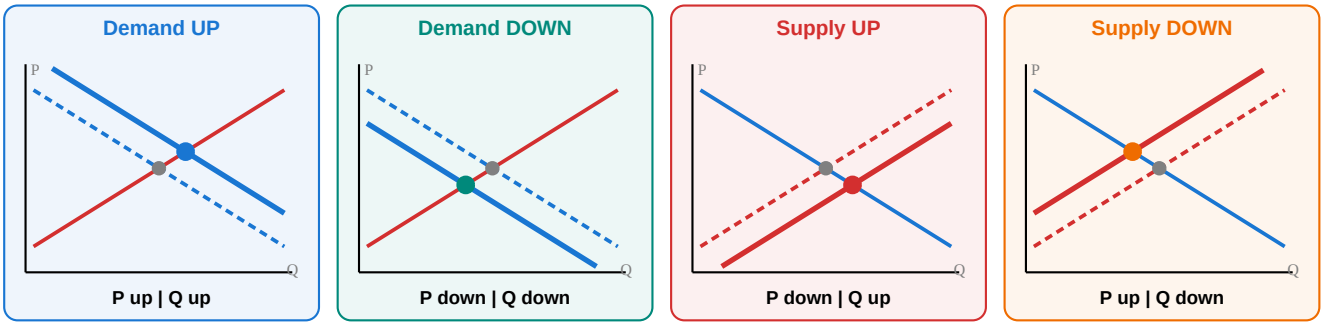
The 3-Step Method and Shifts

STEP 1
Decide which curve shifts: supply, demand, or both

STEP 2
Decide the direction: RIGHT or LEFT

STEP 3
Compare old and new equilibrium: effect on P and Q

FOUR BASIC SHIFTS (grey dot = old E, coloured dot = new E)



MASTER TABLE: EFFECT ON EQUILIBRIUM

Event	Curve shift	Price (P)	Quantity (Q)
Demand increases	D right	Rises	Rises
Demand decreases	D left	Falls	Falls
Supply increases	S right	Falls	Rises
Supply decreases	S left	Rises	Falls
D right AND S right	Both right	**Ambiguous**	Rises
D left AND S left	Both left	**Ambiguous**	Falls
D right AND S left	D right, S left	Rises	**Ambiguous**
D left AND S right	D left, S right	Falls	**Ambiguous**

EXAMPLE 2: Shift in SUPPLY

- **Event:** new technology reduces the cost of producing hybrid cars.
- **Step 1:** S shifts (cost of production). D does not shift.
- **Step 2:** S shifts **RIGHT**.
- **Step 3:** price **falls**, quantity **rises**.

EXAMPLE 3: BOTH shift

- **Events:** gas price rises AND new technology cuts costs.
- **Steps 1-2:** both curves shift RIGHT.
- **Step 3:** Q **rises**; P **ambiguous**.
- If D rises more -> P rises. If S rises more -> P falls.

ACTIVE LEARNING 3: Music Downloads Market

Event	Step 1	Step 2	Step 3 (Result)
A. Price of music CDs falls	D shifts (substitutes)	D LEFT	P falls, Q falls
B. Royalties (cost) fall	S shifts (cost)	S RIGHT	P falls, Q rises
C. Both A and B	Both shift	D left, S right	**P falls**; Q is **ambiguous** (demand fall lowers Q, supply rise raises Q)

KEY RULE: Both curves shift -> ONE of P or Q is certain, the OTHER is **ambiguous**. Always say which shift is bigger decides it.

Maths Problems and Surplus

1. EQUILIBRIUM (Lecture Problem)

Given: Qd = 50 - 5P and Qs = 5 + 10P
Set Qd = Qs: 50 - 5P = 5 + 10P
45 = 15P so **P = 3**
Q = 50 - 5(3) = **35** (check: 5 + 30 = 35)

Surplus at P = 5: Qd = 25, Qs = 55 -> **surplus 30**
Shortage at P = 1: Qd = 45, Qs = 15 -> **shortage 30**
Ceiling at P = 2 (binding): Qd = 40, Qs = 25 -> shortage 15
Floor at P = 4 (binding): Qd = 30, Qs = 45 -> surplus 15

2. SHIFTS AND OTHER PROBLEMS

Problem	Working	Answer
Demand rises to Qd = 65 - 5P (Qs = 5 + 10P)	65 - 5P = 5 + 10P -> 60 = 15P	**P = 4, Q = 45** (both rise)
Supply rises to Qs = 20 + 10P (Qd = 50 - 5P)	50 - 5P = 20 + 10P -> 30 = 15P	**P = 2, Q = 40** (P falls, Q rises)
Qd = 100 - 2P, Qs = 20 + 3P	80 = 5P	**P = 16, Q = 68**
Market demand: Sam 16 - 2P, Dean 8 - P	Add constants and P terms	**Qd = 24 - 3P**
Market supply: Starbucks 3P, Peet's 2P	3P + 2P	**Qs = 5P**
Lattes market: 24 - 3P = 5P	24 = 8P	**P = 3, Q = 15**
PPF: A to C (+200 comp, -100 cars)	100 / 200	**1 computer = 0.5 car**

3. CONSUMER SURPLUS and PRODUCER SURPLUS (Extra Topic)

- **Willingness to pay (WTP):** max price a buyer will pay.
- **Cost:** min price a seller will accept.
- **Consumer surplus (CS)** = WTP - price paid (area below D, above P).
- **Producer surplus (PS)** = price received - cost (area above S, below P).
- **Total surplus** = CS + PS; **maximum at equilibrium** (efficient).
- **Deadweight loss** = fall in total surplus from a distortion.

Buyer	WTP	CS at $5
Anna	$10	5
Ben	$8	3
Chris	$6	1
Total CS		**9**

Seller	Cost	PS at $5
Seller 1	$2	3
Seller 2	$4	1
Seller 3	$5	0
Total PS		**4**

CHECK: Total surplus = 9 + 4 = **13** = (10 + 8 + 6) - (2 + 4 + 5) = 24 - 11. Dina (WTP $4) and Seller 4 (cost $7) do not trade.

Lattes market (Qd = 24 - 3P, Qs = 5P, E: P = 3, Q = 15)	Calculation	Value
Consumer surplus	1/2 x 15 x (8 - 3)	**37.5**
Producer surplus	1/2 x 15 x (3 - 0)	**22.5**
Total surplus	37.5 + 22.5	**60**

Formulas: Equilibrium: Qd = Qs. Surplus = Qs - Qd (price above E). Shortage = Qd - Qs (price below E). Opportunity cost of X = (Y given up) / (X gained). CS (linear) = 1/2 x Q x (choke price - P).

Price Controls, Systems, Exceptions

PRICE CONTROLS (Extra Topic)

Basis	Price CEILING	Price FLOOR
Meaning	Legal **maximum** price	Legal **minimum** price
Binding if set	BELOW equilibrium	ABOVE equilibrium
Result	**SHORTAGE** (queues, black markets)	**SURPLUS** (unsold goods, unemployment)
Example	Rent control, food price caps	Minimum wage, farm support price
Lattes example	Ceiling $2: Qd 18, Qs 10, shortage 8, DWL 6.67	Floor $4: Qd 12, Qs 20, surplus 8, DWL 2.4

ECONOMIC SYSTEMS (Extra Topic)

System	Who decides	Plus	Minus	Example
Market	Prices and private firms	Efficiency, innovation	Inequality	USA
Command	Government planners	Equality, basic needs	Inefficiency, no choice	North Korea
Mixed	Market + government	Balance of both	Govt. inefficiency	Bangladesh, India, UK
Traditional	Custom and habit	Stable, simple	No growth	Tribal groups

EXCEPTIONS TO LAW OF DEMAND

- **Giffen good:** staple inferior good; price up -> demand up (income effect > substitution effect). Example: potatoes in Irish famine.
- **Veblen good:** luxury status good (diamonds, supercars).
- **Expected price rise** or speculative goods (shares, gold).
- **Price as quality signal** when buyers cannot judge quality.

EXCEPTIONS TO LAW OF SUPPLY

- **Perishable goods** (fish, milk): must sell at any price.
- **Fixed or rare supply** (antiques, land): vertical curve.
- **Backward-bending labor supply:** at very high wages, workers choose more leisure.
- **Expectation** of higher future prices: sellers hold back.

FINAL CHEAT SHEET: DIRECTION OF SHIFTS

DEMAND shifts RIGHT when...	SUPPLY shifts RIGHT when...
More buyers | Income up (normal good) | Substitute price up | Complement price down | Tastes favour good | Expect higher future income or prices	Input prices fall | Technology improves | More sellers | (Expect LOWER future price: sellers sell now)

Term	Quick definition
Positive / Normative	What IS (testable) / what SHOULD BE (opinion)
Law of demand / supply	Price up: Qd down / Qs up (other things equal)
Equilibrium	Qs = Qd; surplus above, shortage below
Shift vs movement	Own price = movement along; other factors = shift

BEST OF LUCK FOR YOUR EXAM!

Draw every graph, label every axis, mention "other things equal", and show the 3 steps.